

LOSO Experiment A2

Full sensors (flex+ypr+accel, both hands), EXCLUDING Double_Nothing class

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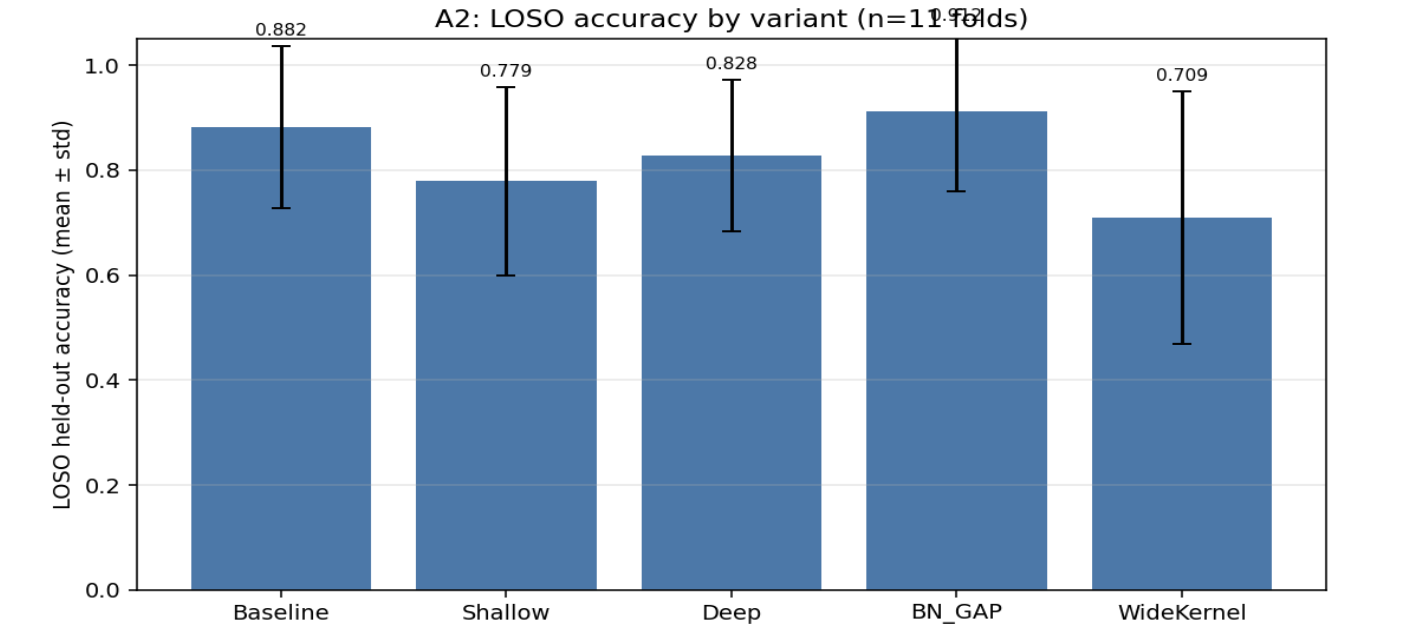
Experiment configuration

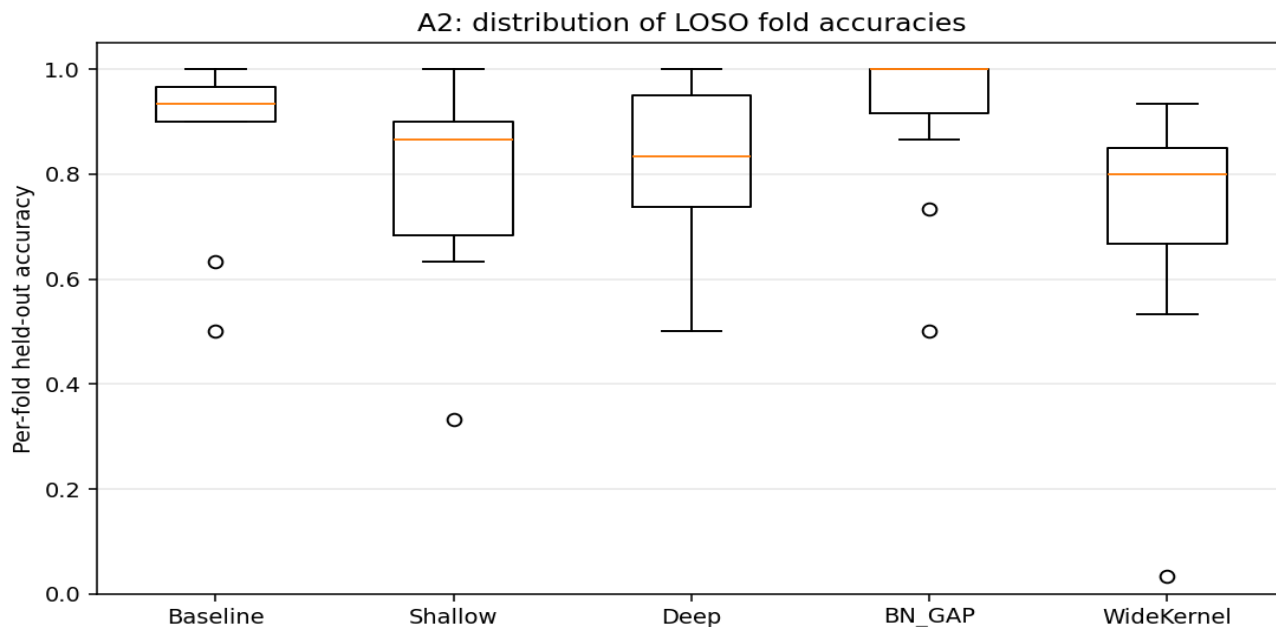
Class set	6 classes excluding Double_Nothing
Classes	Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor modalities	flex, YPR, accel
Hands	left, right
Sensor channels	176
Sequence length	90 steps (resampled)
Filter	Butterworth 10.0Hz order 4
Normalisation	minmax
Augmentation	amplitude_scale + baseline_offset, 3 copies
Inner CV folds	3
Epochs · batch size	8 · 16
LOSO subjects	11: 11_Mansh, 12_Marcus, 14_StephenV2, 15_Tash, 16_Bella, 17_Raquel, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 6_Harry

LOSO summary across all subjects

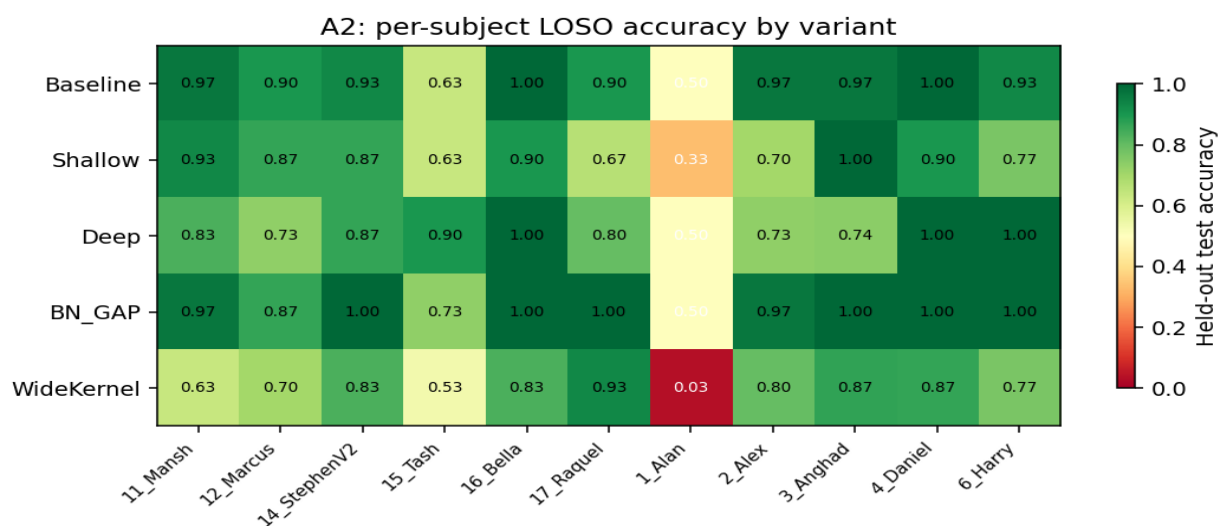
Variant	Params	LOSO mean	LOSO std	Min fold	Max fold	Inner CV mean (avg across folds)
Baseline	113,190	0.8819	0.1547	0.5000	1.0000	0.9765
Shallow	72,454	0.7788	0.1799	0.3333	1.0000	0.8861
Deep	203,878	0.8281	0.1443	0.5000	1.0000	0.9741
BN_GAP	57,190	0.9121	0.1526	0.5000	1.0000	0.9895
WideKernel	131,622	0.7095	0.2407	0.0333	0.9333	0.8891

Best variant by LOSO mean: **BN_GAP**





Per-subject LOSO accuracy heatmap



Per-fold detail (held-out subject = each row)

Held-out subj	Train trials	Baseline	Shallow	Deep	BN_GAP	WideKernel
11_Mansh	301	0.967	0.933	0.833	0.967	0.633
12_Marcus	301	0.900	0.867	0.733	0.867	0.700
14_StephenV2	301	0.933	0.867	0.867	1.000	0.833
15_Tash	301	0.633	0.633	0.900	0.733	0.533
16_Bella	301	1.000	0.900	1.000	1.000	0.833
17_Raquel	301	0.900	0.667	0.800	1.000	0.933
1_Alan	301	0.500	0.333	0.500	0.500	0.033
2_Alex	301	0.967	0.700	0.733	0.967	0.800
3_Anghad	300	0.968	1.000	0.742	1.000	0.871
4_Daniel	301	1.000	0.900	1.000	1.000	0.867
6_Harry	301	0.933	0.767	1.000	1.000	0.767

Variant: Baseline

Params: 113,190

LOSO mean accuracy: 0.8819 ± 0.1547 (min 0.5000, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	4/5	5/5	5/5	5/5	5/5	0.967
12_Marcus	5/5	5/5	5/5	5/5	5/5	2/5	0.900
14_StephenV2	5/5	5/5	5/5	5/5	5/5	3/5	0.933
15_Tash	4/5	5/5	5/5	4/5	0/5	1/5	0.633
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	5/5	5/5	2/5	5/5	5/5	0.900
1_Alan	5/5	5/5	5/5	0/5	0/5	0/5	0.500
2_Alex	5/5	4/5	5/5	5/5	5/5	5/5	0.967
3_Anghad	5/5	5/5	5/5	5/5	6/6	4/5	0.968
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
6_Harry	5/5	5/5	5/5	5/5	3/5	5/5	0.933
MEAN	0.982	0.964	1.000	0.836	0.782	0.727	0.882

Variant: Shallow

Params: 72,454

LOSO mean accuracy: 0.7788 ± 0.1799 (min 0.3333, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	4/5	5/5	5/5	5/5	4/5	5/5	0.933
12_Marcus	5/5	5/5	5/5	1/5	5/5	5/5	0.867
14_StephenV2	2/5	5/5	5/5	4/5	5/5	5/5	0.867
15_Tash	0/5	5/5	5/5	5/5	4/5	0/5	0.633
16_Bella	5/5	5/5	2/5	5/5	5/5	5/5	0.900
17_Raquel	5/5	5/5	5/5	0/5	5/5	0/5	0.667
1_Alan	5/5	0/5	5/5	0/5	0/5	0/5	0.333
2_Alex	3/5	4/5	5/5	0/5	5/5	4/5	0.700
3_Anghad	5/5	5/5	5/5	5/5	6/6	5/5	1.000
4_Daniel	5/5	5/5	2/5	5/5	5/5	5/5	0.900
6_Harry	5/5	4/5	5/5	0/5	4/5	5/5	0.767
MEAN	0.800	0.873	0.891	0.545	0.855	0.709	0.779

Variant: Deep

Params: 203,878

LOSO mean accuracy: 0.8281 ± 0.1443 (min 0.5000, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	0/5	5/5	5/5	5/5	5/5	0.833
12_Marcus	5/5	5/5	5/5	1/5	5/5	1/5	0.733
14_StephenV2	5/5	5/5	1/5	5/5	5/5	5/5	0.867
15_Tash	5/5	5/5	5/5	5/5	2/5	5/5	0.900
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	5/5	5/5	4/5	5/5	0/5	0.800
1_Alan	4/5	0/5	5/5	1/5	5/5	0/5	0.500
2_Alex	5/5	4/5	5/5	5/5	3/5	0/5	0.733
3_Anghad	5/5	5/5	0/5	5/5	6/6	2/5	0.742
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
6_Harry	5/5	5/5	5/5	5/5	5/5	5/5	1.000
MEAN	0.982	0.800	0.836	0.836	0.909	0.600	0.828

Variant: BN_GAP

Params: 57,190

LOSO mean accuracy: 0.9121 ± 0.1526 (min 0.5000, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	4/5	5/5	5/5	5/5	5/5	0.967
12_Marcus	5/5	5/5	5/5	5/5	5/5	1/5	0.867
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	1.000
15_Tash	4/5	5/5	5/5	5/5	2/5	1/5	0.733
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
1_Alan	5/5	5/5	0/5	0/5	5/5	0/5	0.500
2_Alex	5/5	4/5	5/5	5/5	5/5	5/5	0.967
3_Anghad	5/5	5/5	5/5	5/5	6/6	5/5	1.000
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
6_Harry	5/5	5/5	5/5	5/5	5/5	5/5	1.000
MEAN	0.982	0.964	0.909	0.909	0.945	0.764	0.912

Variant: WideKernel

Params: 131,622

LOSO mean accuracy: 0.7095 ± 0.2407 (min 0.0333, max 0.9333)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	0/5	5/5	5/5	1/5	5/5	3/5	0.633
12_Marcus	2/5	5/5	5/5	4/5	5/5	0/5	0.700
14_StephenV2	0/5	5/5	5/5	5/5	5/5	5/5	0.833
15_Tash	0/5	5/5	5/5	1/5	1/5	4/5	0.533
16_Bella	5/5	5/5	5/5	0/5	5/5	5/5	0.833
17_Raquel	5/5	5/5	5/5	5/5	5/5	3/5	0.933
1_Alan	0/5	0/5	1/5	0/5	0/5	0/5	0.033
2_Alex	0/5	4/5	5/5	5/5	5/5	5/5	0.800
3_Anghad	5/5	5/5	4/5	2/5	6/6	5/5	0.871
4_Daniel	4/5	5/5	5/5	2/5	5/5	5/5	0.867
6_Harry	5/5	5/5	5/5	3/5	0/5	5/5	0.767
MEAN	0.473	0.891	0.909	0.509	0.745	0.727	0.709